Banking Amidst Conflict: How Violence Shapes Financial Inclusion in Mexico*

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Abstract

We study how exposure to violence shapes the use of formal financial services in Mexico. Using administrative records on violent incidents, we construct out-of-sample predictions of violence and define violence shocks as unexpected deviations of realized violence from these predictions. We combine these shocks with nationwide administrative transaction data from the National Banking and Securities Commission covering ATM and correspondent banking activity across all municipalities. Measuring financial activity relative to municipality-specific seasonal baselines, we find that unexpected spikes in violence reduce ATM transactions per capita below their norm, with effects concentrated in the month of the shock and the following month and dissipating within two months. In contrast, we find no statistically significant response in correspondent banking activity at any horizon, with point estimates close to zero. This divergence indicates that violence primarily disrupts access to traditional ATM infrastructure rather than inducing broad financial disengagement, and suggests that correspondent banking can serve as a partial buffer against violence-induced disruptions.

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1 Introduction

There is a strong relationship between conflict and poverty: Poverty makes countries more prone to civil war and armed conflict, increasing individuals' exposure to violence, weakening governance and economic performance, thus increasing the risk of conflict (Goodhand, 2001). Examples of countries facing these types of problems include Ethiopia, Indonesia, Uganda, Colombia, Mexico, among others. In consequence, the relationship between conflict and poverty has emerged as a topic of increasing interest among economists and policymakers around the world. According to the World Bank, roughly one-fifth of the global population resides in countries affected by violence or conflict. Even more alarmingly, it is projected that two out of every three individuals living in extreme poverty will live in a country that is affected by violence or conflict (Corral et al., 2020). Empirical research has shown that conflict negatively affects capital (Davis and Weinstein, 2002; Miguel and Roland, 2011), investment (Besley and Mueller, 2012), economic decision-making (Voors et al., 2012), and usage of financial tools (Blumenstock et al., 2021). Conversely, there is a well-established body of literature covering the relationship between financial accessibility and poverty. These studies highlight the positive relationship between financial inclusion and economic growth, household consumption, women's empowerment, self-employment, and financial well-being (Bauchet et al., 2011; Bruhn and Love, 2011; Karlan and Zinman, 2010). However, other studies' findings do not observe a significant impact (Ashraf et al., 2010; Banerjee et al., 2015).

In this paper, we explore the relationship between conflict and the usage of both traditional and non-traditional formal financial services. Specifically, we examine whether an unexpected increase in local violence causes users to pull back from financial activity broadly, or whether its impact is concentrated in particular channels. The distinction matters from policy: if violence causes broad financial disengagement, then insecurity represents a direct barrier to financial inclusion, undermining the reach of both traditional and alternative access infrastructure. If, instead, violence shifts users toward correspondent banking, then expanding the correspondent network could serve as a resilience-building tool in insecure environments.

We exploit two complementary features of Mexico's financial landscape. First, Mexico has a well-documented correspondent banking network centered around OXXO convenience stores, which by 2021 operated more than 20,000 locations in partnership with all major Mexican banks. Banking correspondents are third-party retail establishments such as convenience stores, pharmacies, and supermarkets that are authorized to provide basic financial services including deposits, withdrawals, transfers, and bill payments. OXXO stores are broadly comparable to 7-Eleven in the United States and Carrefour Express formats in Europe in that they are small-format retail locations embedded in neighborhoods and open long hours, but their functional role is broader in the Mexican context. In addition to retail, they serve as key financial access points and operate as a retail-based extension of the formal banking system. This network provides a plausible low-risk alternative to traditional ATM infrastructure, as OXXO stores are integrated into local commercial activity and often perceived as safer environments than standalone ATMs. Second, Mexico experienced substantial variation in violence at the municipal level between 2014 and 2019, driven by cartel competition and broader dynamics linked to the war on drugs. This variation allows us to study the impact of unexpected spikes in violence on financial behavior while controlling for predictable, trend-driven changes in insecurity.

Our identification strategy combines an out-of-sample violence forecasting model with a distributed lag regression framework. We use a long short-term memory (LSTM) neural network to generate municipality-level forecasts of violent incidents. We define a violence shock as any month in which realized violence is at least twice the model's prediction, isolating unexpected deviations from predicted local trends. We then estimate the dynamic response of per-capita financial activity to these shocks, controlling for municipality-specific seasonal patterns and common aggregate time trends.

We use per capita normalization as our primary specification for both ATM and OXXO outcomes. This choice is motivated by the fact that population is mostly exogenous to short-run violence shocks within our four-month analysis window. The number of ATMs and OXXO stores, by contrast, may respond to local security conditions: banks may remove ATMs from dangerous locations, and OXXO stores may temporarily close or reduce hours in response to violence, making per-branch measures endogenous to the shocks we study.

We have two main findings. First, unexpected violence shocks reduce ATM transactions per capita by approximately 1.6 percent in the month of the shock and 1.3 percent the following month, with effects dissipating within two months. A robustness specification using log transactions per ATM yields consistent results, with an effect at lag 2 that is also marginally significant, suggesting a slightly more persistent contraction in machine utilization relative to the per-capita measure. Second, OXXO correspondent banking activity shows no statistically significant response at any horizon, with point estimates an order of magnitude smaller than the ATM effects and mixed in sign. The divergence between channels constitutes our central empirical result: violence disrupts ATM-specific access while leaving correspondent banking unaffected.

The channel asymmetry has a natural economic interpretation. ATMs are unattended machines that signal the presence of cash, making them predictably targeted during periods of heightened insecurity. OXXO stores, by contrast, are staffed retail environments embedded in everyday commercial activity, where the financial transaction is embedded within a broader shopping trip. The presence of other customers and staff provides a natural social deterrent, and the financial component is less visible than at a standalone ATM. These differences imply that violence raises the effective cost of ATM access more sharply than correspondent access, generating the divergence we document.

This paper contributes to three strands of the literature. First, we add to the evidence on how conflict affects financial behavior (Besley and Mueller, 2012; Blumenstock et al., 2024; Boungou et al., 2025; Compaoré et al., 2022; Voors et al., 2012), extending this work to the Latin American context and showing that responses operate through specific transaction channels rather than broad financial disengagement. Second, we use an LSTM walk-forward forecasting approach (Athey and Imbens, 2019; Fischer and Krauss, 2018; Mueller and Rauh, 2018, 2022; Qiu et al., 2020) to construct out-of-sample predictions of violence at the municipal level. Following the shock-identification approaches of Romer and Romer (2004) and Aruoba and Drechsel (2024), we define unexpected violence shocks that complement existing approaches based on plausibly exogenous conflict variation (Blattman and Miguel, 2010). Third, we provide new evidence on the resilience of bank correspondents to violence, contributing to a growing literature on branchless banking and financial access in Latin America (Assunção, 2013; Bruhn and Love, 2014; Carabarín et al., 2018; Célerier and Matray, 2019; Eisele and Villarreal, 2015; Rodrigues-Loureiro et al., 2016). Despite the scale of bank correspondents, particularly in Latin America, their response to security shocks remains largely unexplored.

The paper is organized as follows. Section 2 reviews the role of bank correspondents in the Mexican context. Section 3 describes the data. Section 4 presents the machine learning forecasting approach. Section 5 covers the sample construction and descriptive statistics. Section 6 outlines the empirical strategy. Section 7 presents the results. Section 8 discusses mechanisms and implications. Section 9 concludes.

2 Bank Correspondents in Mexico

Given the lack of penetration of basic financial services in Mexico, both local and federal government have pushed for policies that allow financial institutions to operate through bank correspondents. Bank correspondents are establishments such as retail and convenience stores, supermarkets, pharmacies, postal services, and gas stations, among others that are authorized by financial institutions as a distribution channel and provide access to basic financial services. In other words, bank correspondents are one of the biggest pushes from the private sector and the banking system to increase financial inclusion in the

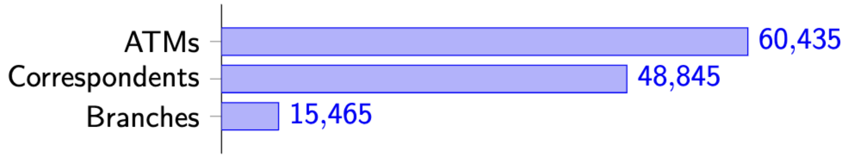


Figure 1: Banking Infrastructure in Mexico (CNBV, 2022)

country.

The regulatory framework that allowed for the operation of bank correspondents in Mexico was approved in December 2008 and bank correspondents were allowed to start operations in November 2009. During the first few years of the policy implementation, banks signed various partnerships with small retail stores and pharmacies to work as bank correspondents; but it wasn't until 2011, when Banco Santander signed a correspondence deal with OXXO (the largest convenience store in Mexico), that the bank correspondents model exploded in Mexico. The deal between Banco Santander and OXXO allowed Santander to immediately reach 325 municipalities with this new business approach (Carabarín et al., 2018). The impact of OXXO joining the bank correspondent model was so big that by early 2021, OXXO had more than 20,000 stores all over Mexico partnering with all major banks in Mexico including BBVA, Citibanamex[1], Santander, Scotiabank, HSBC, Inbursa, Bancoppel, Banregio Afirme, Caja Popular Mexicana, and Hey Banco. By 2011 bank correspondents became a substitute to banking branches for both the supply and demand sides of the market, making it easy for users to do their everyday transactions at a bank correspondent instead of a branch[2].

Despite their rapid expansion and large scale in Latin America's financial inclusion landscape, bank correspondents remain understudied. In Mexico, Eisele and Villareal (2015) found that correspondents entering the market have a positive effect on household income, while Peña and Vázquez (2012) found that correspondents have no significant effect on financial inclusion. In Brazil, Rodrigues-Loureiro et al. (2016) documented a negative relationship between banking branches and correspondent presence, suggesting substitution between channels. These findings highlight how correspondents and branches have become viable, lower-cost options for banks to expand their client base. Yet it remains unclear why a user might prefer a correspondent outlet over a branch or ATM in a given situation. This paper focuses on one specific margin: the role of violence and perceived insecurity in shaping channel choice and overall financial engagement.

[1] The partnership between Citibanamex and OXXO ended in May of 2021.

[2] It is important to note that for urban users, bank correspondents are a substitute to traditional banks, while for rural users, bank correspondents might be the only option.

3 Data

For this project, we draw on four administrative data sources. First, we use administrative records of injuries and deaths from the Mexican Ministry of Health, covering 2012–2019. Second, we use monthly reports from the National Banking and Securities Commission (CNBV; Comisión Nacional Bancaria y de Valores), covering 2014–2019. Third, we use population data from INEGI (Instituto Nacional de Estadística y Geografía, Mexico's national statistics agency) Censuses (2010, 2020) and the 2015 Intercensal Survey. Fourth, we use retail establishment data from the DENUE (Directorio Estadístico Nacional de Unidades Económicas), INEGI's national directory of economic establishments, spanning 2011–2022.

To measure the extent of violence in each municipality, we use administrative records of injuries and deaths classified under the International Classification of Diseases (ICD-10) from Mexico's Ministry of Health from 2012 to 2019. Our measure focuses on incidents consistent with interpersonal aggression and exposure to potentially lethal force. Specifically, we include firearm assaults; attacks with sharp objects such as knives; assaults with blunt objects or physical force; and severe forms of aggression including asphyxiation, strangulation, and drowning. We also include assaults involving hazardous materials. To reduce measurement error from classification challenges, we additionally incorporate injuries caused by firearms, explosives, or sharp objects with undetermined intent. We aggregate events to the municipal level to construct a measure of the overall security environment faced by households in that jurisdiction.

The banking data come from monthly CNBV reports, self-reported by banking institutions at the branch level. These monthly reports include the number of ATMs, the number of ATM transactions, the number of correspondent users, total transactions at correspondent locations, and the total amount transacted in pesos. All banking data are aggregated from the branch level to the municipality level. An important limitation is that for ATMs we observe only the *count* of transactions, not the peso value of individual withdrawals; for OXXO correspondents we observe both the count and the total peso volume of transactions, but not the distribution of amounts across individual transactions.[3] The banking regression sample covers January 2014 to December 2019. Note that the sample ends in 2019 deliberately: extending it into 2020 would require grappling with the onset of the COVID-19 pandemic, which caused unprecedented disruptions to in-person financial activity beginning in March 2020 and would confound our violence shock identification.

[3] This asymmetry reflects what banks self-report to the CNBV. For ATMs, only transaction counts are required; for correspondents, the full peso volume is also reported. Our analysis therefore focuses on transaction counts, which are available consistently for both channels.

To construct per capita measures of financial activity, we use municipality-level population data from the Population and Housing Censuses and the Intercensal Survey conducted by INEGI. INEGI conducts a full population and housing census approximately every ten years and an intercensal survey in between, yielding population counts at roughly five-year intervals. The most recent rounds used in our analysis correspond to the 2010 Census, the 2015 Intercensal Survey, and the 2020 Census, providing benchmark population figures for each municipality at three points in time.

Because our panel spans 2012 to 2019 on a monthly basis, we linearly interpolate between the 2010 and 2015 benchmarks for years 2011–2014 and between the 2015 and 2020[4] benchmarks for years 2016–2019. This procedure yields a municipality-year population series that evolves smoothly between survey rounds while anchoring to the officially measured population at each benchmark year. The interpolated annual estimates are then assigned uniformly across calendar months within each year, so that population in municipality m in month t equals the interpolated annual estimate for the year containing t. This approach is standard in the literature when census data are available only at multi-year intervals.

To measure the geographic presence of OXXO convenience stores, we use establishment-level data from the *Directorio Estadístico Nacional de Unidades Económicas* (DENUE), produced by the Instituto Nacional de Estadística y Geografía (INEGI). DENUE is a census-style directory of formally registered economic establishments in Mexico and is updated periodically. Each observation corresponds to a single physical establishment and includes the establishment name, legal entity name, six-digit industry classification (SCIAN), municipality and state identifiers, geographic coordinates, and registration date. DENUE is establishment-level rather than firm-level; each OXXO store appears as a separate observation.

We construct a municipality-by-year panel of OXXO stores by stacking historical DENUE editions between 2011 and 2022. For each edition, we identify OXXO establishments using the establishment name field, flagging observations whose name contains the string "OXXO". We then collapse the data to the municipality level and count the total number of OXXO establishments in each municipality-year. For years in which multiple DENUE editions are available, we use the edition closest to year-end to approximate annual counts.

This procedure yields a municipality-year panel capturing the stock of OXXO stores present in each municipality at a given point in time. As a validation check, we aggregate municipality counts to the national level and verify that trends closely track publicly reported total OXXO store counts over the same period.

[4]The Census of Population and Housing 2020 was carried out from March 2nd to the 27th (INEGI). According to the World Health Organization, Mexico entered Phase 2 of the coronavirus pandemic on March 23 with 367 confirmed cases. Thus, data collection was largely complete before the pandemic affected normal activity in Mexico.

4 Forecasting Methodology for Municipal Violence in Mexico

As mentioned in Section 3, our measure of violence focuses on incidents consistent with interpersonal aggression and exposure to lethal force, drawn from ICD-10 administrative records. The goal is to capture variation in the local security environment (via exposure to violence) rather than isolated accidental events. Taken together, this approach prioritizes acts that reflect interpersonal violence while excluding clearly accidental injuries, with the goal of making this measure a consistent proxy for local victimization risk.

Our forecasting exercise operates on a municipality-by-month panel, where the predicted variable x is a monthly count of violent incidents at the municipality level. We generate forecasts using a walk-forward design: for each forecast year $t \in \{2014, \ldots, 2019\}$, we train on data from January 2012 through December of year $t-1$ and predict all months in year t. The model is estimated separately for each municipality using only its own historical data. We use a long short-term memory (LSTM) network for forecasting, and define violence shocks as the residuals between realized and predicted values. These residuals are then used to identify the effect of local insecurity on banking activity.

We use an LSTM rather than more standard time-series or machine learning models because municipal violence in Mexico exhibits nonlinear dynamics that depend on the ordering of past events. Relative to ARIMA-type models, which impose fixed lag structures, or random forests, which require pre-specified lag features (Fischer and Krauss, 2018), LSTMs learn directly from the data to understand which lags matter and how to weight them without requiring the researcher to pre-specify lag structure. This is particularly useful in our setting, where violence is intermittent and highly state-dependent.

Fischer and Krauss (2018) show in a large-scale financial forecasting setting that LSTMs outperform random forests, feedforward neural networks, and logistic regression in out-of-sample prediction. Qiu et al. (2020) further document that gated recurrent architectures perform well in noisy sequential environments, which is directly relevant here given the volatility of monthly violence counts.

Our identification strategy follows the logic of Romer and Romer (2004). Just as they define monetary policy shocks as deviations of realized policy from central bank internal forecasts, we define violence shocks as deviations of realized violence from out-of-sample LSTM predictions. Aruoba and Drechsel (2024) extend this approach by replacing the Fed's institutional forecasts with machine-learning predictions to construct monetary policy shocks, offering a direct methodological parallel to our use of LSTM predictions as the main tool for identification. In all cases, the forecast is constructed using only information available prior to the period in question, so the residual isolates unexpected variation relative to the model's training data.

Formally, our walk-forward design guarantees this property: for any forecast-year municipality-month (m, t), the LSTM prediction $\widehat{\text{Violence}}_{mt}$ is computed from training data ending at December $t-1$. No information from period t or later enters the model. The shock indicator ($\text{Shock}_{mt} = \mathbf{1}[\text{Violence}_{mt}/\widehat{\text{Violence}}_{mt} \geq 2]$) therefore captures purely unanticipated violence spikes — events that were not expected given the municipality's own history. Under the assumption that agents' financial behavior responds to the same information the LSTM conditions on, the shock indicator is exogenous to contemporaneous financial decisions.

This design addresses two main concerns. First, it removes predictable trends in violence: municipalities with rising violence paths will have higher predicted values, so only deviations from those paths are classified as shocks. Second, it accounts for municipality-specific seasonality and persistence patterns, so that systematic monthly variation does not mechanically generate shocks. The decline in shock incidence from 7.42 percent in 2014 to 4.30 percent in 2019 (Table 2) is consistent with improved predictive performance as the model accumulates more training data over time.

5 Sample Construction and Descriptive Statistics

For the analysis, we include only municipalities with at least one observed transaction in that channel during the study period. This selection reflects having some formal financial presence rather than none; municipalities with no ATMs and no OXXO stores are not informative for our analysis of channel-specific responses to violence.[5]

Our primary outcome variables are the logarithm of per-capita transaction counts:

$$Y_{mt}^{\text{ATM}} = \log\left(\frac{\text{ATM transactions}_{mt}}{\text{Population}_{mt}}\right) \tag{1}$$

$$Y_{mt}^{\text{OXXO}} = \log\left(\frac{\text{OXXO transactions}_{mt}}{\text{Population}_{mt}}\right) \tag{2}$$

We use per-capita outcomes as the primary specification because branch and store counts may respond endogenously to violence, making per-branch denominators endogenous. In other words, banks may

[5] A concern is that the municipalities excluded from each panel may be systematically different from those included. In general, municipalities lacking ATMs tend to be smaller and more rural; municipalities lacking OXXO stores tend to be those in which OXXO had not yet expanded. We address the concern that compositional differences between the ATM and OXXO panels drive our results directly in Section 7.1.1, where we restrict both samples to municipalities that appear in both panels simultaneously and find results that are, if anything, stronger.

relocate ATMs away from high-violence areas, and stores may temporarily close. Population, by contrast, is exogenous at short horizons. As robustness checks, we also report results for log transactions per ATM and log transactions per OXXO store.

Table 1 reports the summary statistics for the main variables. Panel A shows 165,384 municipality-month observations across 2,297 municipalities over the 2014–2019 period. On average, a municipality records 1.08 violent incidents per month, though the distribution is right-skewed. The LSTM-predicted violence series averages 1.71 incidents per municipality-month, slightly higher than the realized mean. Across all municipality-months, 5.6 percent are classified as experiencing an unexpected violence shock, defined as realized violence at least twice the LSTM-predicted level.

The ATM regression sample covers 96,659 municipality-month observations across 1,516 municipalities with at least one ATM. Monthly ATM transactions average 121,374 per municipality, with substantial dispersion driven by large urban areas. On a per capita basis, the average municipality processes 0.76 ATM transactions per resident per month.

The OXXO correspondent sample is more geographically concentrated, covering 60,710 municipality-month observations across 1,016 municipalities with at least one OXXO store. The average municipality in this sample hosts 14 OXXO stores and has a population of 118,890, relatively larger than the ATM sample, reflecting OXXO's concentration in higher-density areas. Also, OXXO transactions per capita are right-skewed, driven by a small number of large urban municipalities (e.g. Mexico City and its metropolitan area) characterized by dense OXXO coverage and high transaction volumes.

6 Empirical Strategy

6.1 Violence Shocks

To capture unexpected surges in violence, we construct a measure of violence shocks based on deviations from predicted levels. As mentioned in Section 3, we measure violence using the number of violence-related injuries and deaths at the municipality-month level. Specifically, we first define a shock index as the ratio of realized to predicted violence in municipality m at time t:

$$\text{ShockIndex}_{mt} = \frac{\text{Violence}_{mt}}{\widehat{\text{Violence}}_{mt}}, \tag{3}$$

where $\widehat{\text{Violence}}_{mt}$ denotes the LSTM-predicted level of violence.

We then define a binary indicator for a violence shock. A municipality is classified as experiencing a shock in month t if realized violence is at least twice the predicted level:

$$\text{Shock}_{mt} = \mathbf{1}\left[\text{ShockIndex}_{mt} \geq 2\right]. \tag{4}$$

The threshold of two is chosen to capture extreme realizations of violence that substantially exceed predicted levels, and to distinguish large shocks from routine variation. Robustness checks at alternative thresholds of 1.5× and 1.75× the prediction yield almost identical results. By using a threshold based on relative deviations rather than absolute levels, the measure is comparable across municipalities with different baseline levels of violence. This definition of shocks is used throughout the empirical analysis.

Table 2 reports the incidence of violence shocks by year, where a shock occurs when realized violence is at least twice the predicted level. Over the full sample, 9,198 of 165,384 municipality-months are classified as shocked. Annual shock rates decline over the period which is consistent with the walk-forward design of our forecasting exercise: as the LSTM is trained with more data, its predictions become more accurate over time. Across all six years, 2,038 of 2,297 municipalities (about 89 percent) experience at least one shocked month, confirming that unexpected violence is not concentrated in a small subset of municipalities.

6.2 Empirical Framework

To examine the dynamic effects of unexpected violence on local financial activity, we estimate the following distributed lag specification:

$$Y_{mt} = \sum_{i=0}^{4} \beta_i \, \text{Shock}_{m,t-i} + \alpha_{mc} + \lambda_{yt} + \varepsilon_{mt}, \tag{5}$$

where m indexes municipalities, t indexes months, c denotes the calendar month (1–12), and y denotes years. Municipality×calendar-month fixed effects (α_{mc}) absorb each municipality's average outcome level for a given calendar month, removing both permanent cross-municipality differences and recurring seasonal patterns. Year×month fixed effects (λ_{yt}) absorb aggregate shocks common to all municipalities in a given year-month, including macroeconomic fluctuations and national time trends. Standard errors are clustered at the municipality level.

Our identification relies on two core assumptions. First, the key identifying assumption is that, conditional on municipality×calendar-month and year×month fixed effects, the violence shock indicator Shock_{mt} is as good as randomly assigned. Because the municipality×calendar-month fixed effects (α_{mc})

absorbs the typical transaction level for each municipality-season cell, the design compares financial activity during a shock month relative to the same municipality's activity in the same season in other years. The identifying variation is thus within-municipality and within-season deviation from baseline, induced by a violence shock that the LSTM model did not anticipate. Under this assumption, our coefficients of interest (β_i) are the causal effect of an unexpected violence shock on log financial activity at lag i.

Second, the exclusion restriction requires that violence shocks affect financial behavior only through the security channel. A potential threat would be if municipalities that experience unexpected violence spikes also simultaneously experience correlated negative economic shocks that independently reduce financial activity. We address this by adding the year$\times$month fixed effect to absorb any shock common across municipalities in a given month.

6.3 Pre-Trends Test

We formally test for pre-existing differential trends by augmenting the main specification with four leads of the shock indicator, that is, future shocks relative to the outcome period, alongside the existing four lags. We report the estimated lead coefficients together with the lag coefficients in event-study form. Under the null of no pre-trend, all four lead coefficients should be indistinguishable from zero.

The pre-period coefficients at $t-4$ and $t-3$ are close to zero and statistically insignificant across all four outcomes. The coefficient at $t-2$ is modestly negative and marginally significant for ATM outcomes, around -1.0%, but there is no corresponding effect for OXXO outcomes. A confounding economic pre-trend would be expected to reduce activity in both channels at the same time. The fact that this pattern appears only in the ATM channel is not consistent with that interpretation. Instead, it is more likely capturing the tendency of severe violence episodes to cluster across adjacent months rather than a pre-existing economic decline.

We also test the joint null that all four lead coefficients are equal to zero using an F-test. We fail to reject this null across all four outcomes under the main shock definition, defined as at least twice predicted violence. The p-values are 0.20 for log ATM transactions per capita, 0.39 for log ATM transactions per ATM, 0.60 for log OXXO transactions per capita, and 0.21 for log OXXO transactions per store. These results are robust to restricting the sample to the common-municipality subsample—municipalities that appear in both the ATM and OXXO panels—where all four joint F-tests continue to fail to reject the null ($p = 0.25$, 0.47, 0.31, 0.21, respectively). Taken together, these results, along with the LSTM-based identification that conditions on municipality-specific violence histories, support a causal interpretation of the estimated effects.

6.4 Placebo Test

We define a placebo shock by lowering the threshold of the shock indicator in equation 4, $\text{Shock}_{mt} = \mathbf{1}[\text{ShockIndex}_{mt} \leq 0.5]$, restricting attention to municipality-months with at least one violence-related death. This captures unexpectedly peaceful conditions relative to the forecast.

The intuition is straightforward. If LSTM forecast errors alone were driving the main results then the new definition of the shock ($\text{Shock}_{mt} = \mathbf{1}[\text{ShockIndex}_{mt} \leq 0.5]$) should produce effects of similar sign and magnitude. If, instead, the mechanism operates through physical disruption from elevated violence, the placebo should yield null results.

We find the latter. All five distributed-lag coefficients on $\text{Shock}_{\text{low}}$ are close to zero and statistically insignificant across all four outcomes (Appendix Table A5), with no coefficient exceeding 0.7% in magnitude. These results provide no evidence against the identification strategy.

7 Results

Table 3 reports the effect of unexpected violence shocks on ATM transactions per capita relative to municipalities' historical norms. The distributed lag specification includes the contemporaneous shock and four monthly lags, with municipality and time fixed effects throughout. In other words, the model traces how ATM usage within a municipality evolves over the five months following an unexpected violence shock, relative to that municipality's baseline trend and common time-varying factors.

ATM transactions per capita fall by 1.6 percent in the month of the shock and by 1.3 percent the month after the shock. This suggests a short-run contraction in aggregate ATM usage, concentrated in the first two months following the shock, consistent with the idea of temporary mobility frictions or risk-avoidance behavior rather than persistent financial disengagement.

We also present robustness results using alternative thresholds for the violence shock definition (Table A3) using transactions per ATM in (Table A1). Results across thresholds are consistent with the baseline per-capita specification in both timing and magnitude under the 2× definition of a violence shock. The per-ATM specification, however, shows a slightly more persistent contraction in machine utilization. One possible explanation is a supply-side adjustment: if some ATMs temporarily close during high-violence months, the denominator shrinks while transactions take longer to recover.

Table 4 presents our key result for correspondent banking: the effect of violence shocks on OXXO transactions per capita relative to each municipality's historical baseline.

OXXO transactions per capita do not fall below their historical baseline following a violence shock. Point estimates are near zero across all five lags and none is statistically significant. These results contrast with the ATM per capita findings, where effects are significant in the month of the shock and the month after.

Analogous to the ATM analysis, we present robustness results using alternative thresholds for the violence shock definition (Table A4) and an alternative outcome measure—transactions per OXXO store (Table A2). Both the alternative thresholds and the per-store specification yield estimates that are consistent with the per-capita results: point estimates are close to zero and statistically insignificant.

Comparing the two channels, a clear asymmetry emerges: violence shocks reduce ATM activity but leave correspondent banking transactions unchanged.

The divergence between ATM and OXXO responses constitutes our central empirical finding. In the month of a shock, ATM transactions per capita fall by 1.6 percent while OXXO transactions show no detectable change. This pattern is inconsistent with broad financial disengagement (which would predict negative responses across both channels) and inconsistent with substitution (which would predict an increase in OXXO activity when ATM activity falls). Instead, the evidence points to channel-specific ATM access disruption that leaves correspondent banking unaffected.

7.1 Robustness Checks

7.1.1 Common-Municipality Sample

A potential concern with the baseline results is that the ATM and OXXO panels cover different sets of municipalities. Municipalities in the ATM sample but not the OXXO sample may be systematically different (perhaps more urban, with higher baseline financial activity), and this compositional difference could in principle drive the divergence we observe. To address this, we re-estimate our main specification (equation 5) restricting both samples to the set of municipalities that appear with positive transactions in both the ATM and OXXO panels simultaneously.

Tables 5 and 6 report the results. The common sample contains 966 municipalities, representing 95 percent of the OXXO panel (1,016 municipalities) and 64 percent of the ATM panel (1,516 municipalities). In municipality-month terms, the ATM per-capita outcome covers 61,058 observations and the OXXO per-capita outcome covers 53,537, compared to 87,962 and 55,378 in the full samples. The restriction essentially eliminates municipalities with ATMs but no OXXO presence. This condition reduces a larger share of ATM observations while the OXXO sample shrinks only modestly, confirming that OXXO stores are predominantly located in municipalities that also have ATMs.

The findings are robust and, if anything, sharpen in this common sample. ATM transactions per capita fall by 1.8 percent in the month of the shock, compared to 1.6 percent in the full sample. Similarly, the per-ATM measure shows an even larger contemporaneous impact of 1.9 percent, with an effect of 1.2 percent in the second lag. Just as in the full sample, OXXO outcomes remain entirely flat: all point estimates are close to zero and none approach statistical significance. The divergence between channels therefore cannot be attributed to comparing inherently different types of municipalities. Within municipalities served by both ATMs and OXXO correspondents, unexpected violence suppresses ATM usage while leaving correspondent banking unaffected.

7.1.2 Violence Shock Threshold

A natural concern is whether the results are sensitive to the choice of threshold used to define an unexpected violence shock. The baseline specification classifies a municipality-month as shocked when realized violence exceeds twice the LSTM forecast ($2\times$). We assess robustness by re-estimating the full model at thresholds of $1.5\times$ and $1.75\times$ the prediction. These lower thresholds classify more municipality-months as shocked and therefore capture a broader set of violence events, including moderate surges in addition to extreme spikes.

Table A3 and Table A4 report the joint all-lags specification across all three thresholds for all four outcomes: ATM transactions per capita, ATM transactions per ATM, OXXO transactions per capita, and OXXO transactions per store. Four patterns emerge consistently.

First, the ATM results hold at every threshold regardless of how the outcome is scaled. Both the per-capita and per-ATM specifications show negative, statistically significant coefficients at t and $t+1$ across all three thresholds. Second, the per-ATM specification at the $2\times$ threshold reveals a significant effect extending to lag 2 (two months after the shock), suggesting that the most severe shocks depress transaction intensity at individual machines for a slightly longer window. Third, effects concentrate in months t and $t+1$ and attenuate toward zero by $t+4$, regardless of threshold. Fourth, both OXXO outcomes — per capita and per store — are uniformly small and statistically insignificant across all three thresholds. The ATM–OXXO divergence is therefore not an artifact of the threshold definition or the choice of scaling. Together, these results confirm that the main findings are robust to the definition of the shock variable.

Overall, the robustness exercises point to the same conclusion. Restricting attention to municipalities served by both channels, varying the violence-shock threshold, and scaling transactions by population or service points all yield qualitatively similar results. Violence shocks consistently reduce ATM activity while leaving correspondent banking transactions largely unchanged. This stability across alternative samples, thresholds, and outcome definitions increases confidence that the observed divergence reflects a genuine difference in how these financial access channels respond to local violence.

8 Discussion

The ATM contraction is concentrated in the two months immediately following an unexpected violence spike and dissipates entirely within three months. This timing is consistent with temporary risk avoidance. ATMs are unattended machines that signal the presence of cash, making them predictably targeted during periods of heightened insecurity. When violence spikes unexpectedly, households may defer cash withdrawals because approaching an ATM is perceived as physically dangerous, or because disrupted mobility reduces cash demand. Both channels predict a short-run reduction that recovers as conditions normalize. The per-ATM result (Appendix Table A1) adds nuance: the effect at lag 2 is marginally significant, suggesting that machine utilization recovers slightly more slowly than aggregate per-capita demand. This is consistent with a partial supply-side adjustment, as some ATMs may be temporarily removed or relocated in the immediate aftermath of a shock, taking one additional month to unwind relative to the demand-side response.

The absence of a correspondent banking response has a clear behavioral interpretation. OXXO stores are staffed retail environments where financial transactions are embedded within everyday commercial activity. The presence of other customers and store staff provides a natural deterrent against robbery, and the financial component of the visit is less visible than at a standalone ATM. Moreover, the regular trip to the corner OXXO store is less discretionary than a trip to an ATM: users conducting bill payments, purchasing airtime, or sending small remittances may be less willing to defer these transactions than users seeking cash withdrawals. The per-store result (Table A2) reinforces this interpretation: the positive though insignificant per-store coefficients suggest that stores which remain open may absorb displaced demand, with individual outlets seeing slightly more activity per location. This is consistent with the OXXO network functioning as a resilient financial infrastructure precisely because its transaction environment is inherently more secure.

These findings carry two key policy implications. First, in high-violence settings, expanding the ATM network alone may not be sufficient to sustain financial inclusion if violence periodically suppresses ATM demand. The transient nature of the effect, with full recovery within three months, suggests a behavioral rather than structural barrier, but even temporary reductions in financial engagement may have lasting consequences for savings accumulation, credit access, and household welfare in settings where financial market participation is already fragile. Second, the OXXO correspondent network is genuinely resilient to violence shocks. Policymakers seeking to maintain financial access in high-violence municipalities should consider correspondents as a complementary infrastructure that continues to function when traditional ATM usage is suppressed. Expanding correspondent coverage, particularly in municipalities with high baseline violence or frequent unexpected shocks, may reduce the aggregate welfare cost of the ATM-specific disruptions we document.

9 Conclusion

We study how unexpected violence shocks shape the use of formal financial services in Mexico using administrative data on ATM transactions and OXXO correspondent banking activity across all municipalities from 2014 to 2019. We combine a municipality-level LSTM forecasting model with a distributed lag regression framework, absorbing municipality-specific seasonal baselines and aggregate time trends. Our main finding is that violence shocks generate a short-run ATM-specific access disruption while leaving OXXO correspondent banking unaffected. ATM transactions per capita fall by 1.6 percent in the month of the shock and 1.3 percent the following month; effects dissipate within two months. On the other hand, OXXO correspondent activity shows no statistically significant response at any horizon, with point estimates an order of magnitude smaller than the ATM effects. The divergence between channels is inconsistent with both broad financial disengagement and a simple substitution story across banking channels.

The mechanism we propose is channel-specific risk avoidance. ATMs, as unattended cash-signaling infrastructure, become more dangerous to approach when violence spikes unexpectedly. OXXO stores, as staffed, socially embedded retail environments, do not face the same increase in perceived risk, and their transactions may be less discretionary and more routine. The result is a temporary, channel-specific contraction in ATM usage with no parallel decline in correspondent activity.

From a policy standpoint, correspondent banking emerges as a genuine buffer against violence-induced financial disruptions. Expanding correspondent coverage in high-violence municipalities may reduce the welfare costs associated with the ATM-specific disruptions we document. Because the OXXO correspondent network functions as a resilient alternative even during acute insecurity episodes, policies that promote correspondent expansion in underserved municipalities may yield dual benefits: broader baseline financial inclusion and greater resilience to security shocks.

Future work should examine heterogeneity across the distribution of insecurity, distinguishing between municipalities with endemic violence and those experiencing acute spikes, quantify the extent of substitution from ATMs to correspondent banking, and explore the role of mobile and digital banking as further buffers against physical access frictions in violent settings.

10 Tables

Table 1: Summary Statistics

	Mean	SD	P25	P75	*N*
Panel A: Violence and Shock Variables (2,297 municipalities; 165,384 municipality-months)					
Violent incidents (actual, monthly)	1.08	5.31	0.00	1.00	165,384
Violent incidents (predicted, monthly)	1.71	3.59	0.43	1.60	165,384
Panel B: ATM Variables (1,516 municipalities; 96,659 municipality-months)					
ATM transactions (monthly total)	121,374	412,002	3,618	48,994	96,659
Number of ATMs	35.46	128.48	1.00	13.00	96,659
Municipality population	84,617	183,926	15,473	65,313	96,606
ATM transactions per capita	0.760	0.918	0.208	0.997	96,606
Panel C: OXXO Correspondent Variables (1,016 municipalities; 60,710 municipality-months)					
OXXO transactions (monthly total)	3,480,329	24,800,000	1,354	39,440	60,710
Number of OXXO stores	14.02	37.35	1.00	8.00	60,710
Municipality population	118,890	223,368	21,546	101,148	60,694
OXXO transactions per capita	25.27	82.87	0.045	0.266	60,694

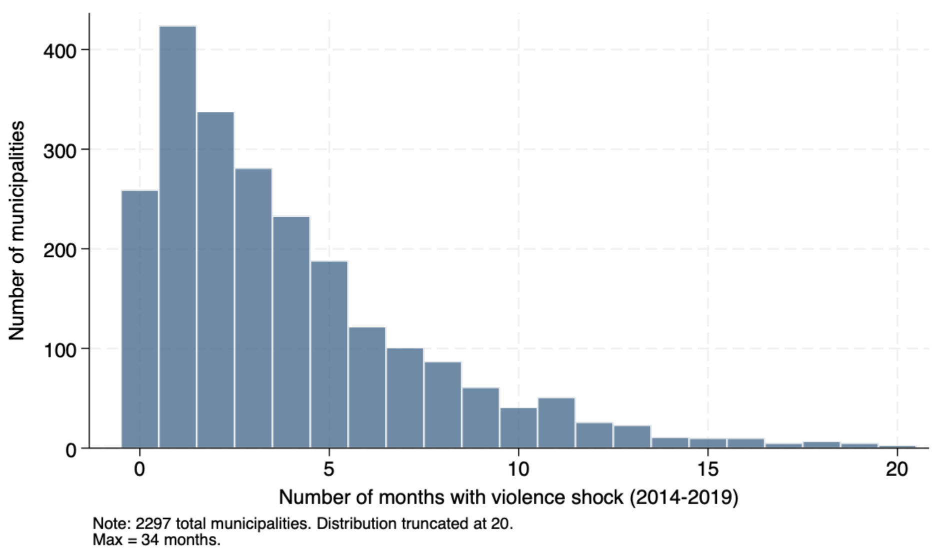
Notes: Unit of observation is a municipality × month. Sample period is 2014–2019. The ATM and OXXO samples are restricted to municipalities with at least one ATM or OXXO store, respectively. Municipality population is linearly interpolated between the 2010, 2015, and 2020 INEGI censuses and assigned uniformly across months within each year.

Table 2: Violence Shock Incidence by Year, 2014–2019

(1) Year	(2) Municipalities	(3) Muni-Months Observations	(4) Muni-Months Shocked	(5) Share of Muni-Months Shocked (%)	(6) Municipalities with $\geq$ 1 Shock
2014	2,297	27,564	2,044	7.42	1,097
2015	2,297	27,564	1,553	5.63	886
2016	2,297	27,564	1,475	5.35	833
2017	2,297	27,564	1,608	5.83	827
2018	2,297	27,564	1,332	4.83	754
2019	2,297	27,564	1,186	4.30	702
Total	2,297	165,384	9,198	5.56	2,038

Notes: A municipality-month is classified as a shock if realized violence is at least twice the LSTM-predicted level. Column (3) reports total municipality-month observations; column (4) reports those with a shock; column (5) reports the corresponding percentage; and column (6) reports municipalities experiencing at least one shock in the given year. The total row reports those over 2014–2019.

Figure 2: Distribution of Violence Shock Months Across Municipalities, 2014–2019



Notes: The figure shows the distribution of the number of months classified as experiencing a violence shock for each of the 2,297 municipalities over the 2014–2019 period. A municipality-month is classified as a shock if realized violence is at least twice the LSTM-predicted level. The vertical axis reports the number of municipalities in each bin.

Figure 3: Average Monthly Violence-Related Rate by Municipality, 2014–2019

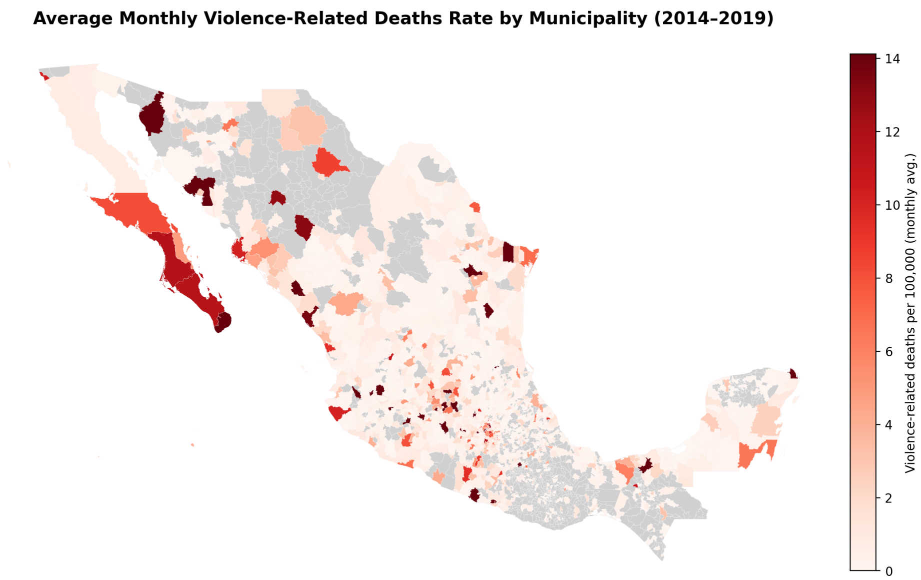
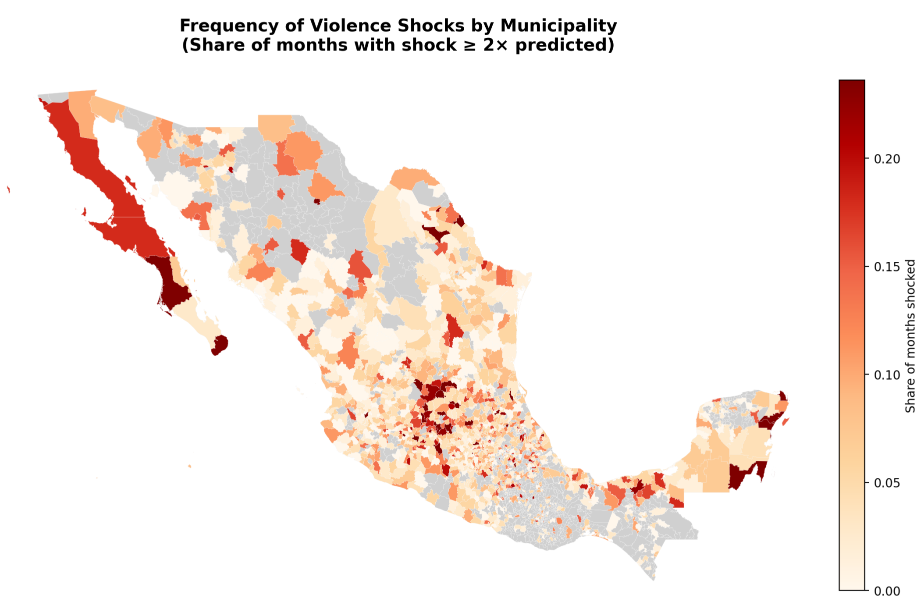


Figure 4: Frequency of Violence Shock (2x) by Municipality, 2014–2019



10.1 ATM Per Capita Results

Table 3: Effect of Violence Shocks on Log ATM Transactions Per Capita

	Log ATM Transactions p.c.
Shock_t	-0.0159^{***}
	(0.00580)
Shock_{t-1}	-0.0129^{**}
	(0.00536)
Shock_{t-2}	-0.00837
	(0.00546)
Shock_{t-3}	-0.00566
	(0.00562)
Shock_{t-4}	-0.00121
	(0.00542)
Municipality×Calendar-Month FE	Yes
Year×Month FE	Yes
N	87,962

The dependent variable is the log of monthly ATM transactions per capita. A municipality-month is classified as a shock if realized violence is at least twice the LSTM-predicted level. Shock_{t-k} denotes a violence shock occurring k months prior to period t, so that the coefficients capture the effect of a shock on the outcome up to k months after its occurrence. All five lags (from t to $t-4$) are included simultaneously in a distributed lag specification. Coefficients approximate percentage changes in ATM transactions per capita. Standard errors are clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

10.2 OXXO Transactions Per Capita Results

Table 4: Effect of Violence Shocks on Log OXXO Transactions Per Capita

	Log OXXO Transactions p.c.
Shock_t	-0.00153
	(0.00534)
Shock_{t-1}	-0.0000192
	(0.00486)
Shock_{t-2}	-0.00113
	(0.00476)
Shock_{t-3}	-0.00141
	(0.00472)
Shock_{t-4}	-0.000358
	(0.00497)
Municipality×Calendar-Month FE	Yes
Year×Month FE	Yes
N	55,378

The dependent variable is the log of monthly OXXO transactions per capita. A municipality-month is classified as a shock if realized violence is at least twice the LSTM-predicted level. Shock_{t-k} denotes a violence shock occurring k months prior to period t, so that the coefficients capture the effect of a shock on the outcome up to k months after its occurrence. All five lags (from t to $t-4$) are included simultaneously in a distributed lag specification. Coefficients approximate percentage changes in OXXO transactions per capita. Standard errors are clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ATM Results — Common-Municipality Sample

Table 5: Effect of Violence Shocks on ATM Outcomes — Common-Municipality Sample

	Log ATM Trans. p.c.	Log ATM Trans. per ATM
Shock_t	-0.0182^{***}	-0.0196^{***}
	(0.00615)	(0.00617)
Shock_{t-1}	-0.00991^{*}	-0.00837
	(0.00550)	(0.00549)
Shock_{t-2}	-0.00909	-0.0123^{**}
	(0.00566)	(0.00599)
Shock_{t-3}	-0.00265	-0.00421
	(0.00557)	(0.00575)
Shock_{t-4}	-0.00536	-0.00760
	(0.00608)	(0.00626)
Mun × Cal-Month FE	Yes	Yes
Year × Month FE	Yes	Yes
N	61,058	61,100

Sample restricted to municipalities with positive transactions in both the ATM and OXXO panels. All five lags included simultaneously. SE clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OXXO Results — Common-Municipality Sample

Table 6: Effect of Violence Shocks on OXXO Outcomes — Common-Municipality Sample

	Log OXXO Trans. p.c.	Log OXXO Trans. per Store
Shock_t	-0.000228	0.00334
	(0.00543)	(0.00623)
Shock_{t-1}	0.000876	0.00366
	(0.00494)	(0.00595)
Shock_{t-2}	-0.000138	0.00405
	(0.00484)	(0.00554)
Shock_{t-3}	-0.000315	0.00394
	(0.00478)	(0.00598)
Shock_{t-4}	0.000398	0.00731
	(0.00506)	(0.00617)
Mun $\times$ Cal-Month FE	Yes	Yes
Year $\times$ Month FE	Yes	Yes
N	53,537	48,260

Sample restricted to municipalities with positive transactions in both the ATM and OXXO panels. All five lags included simultaneously. SE clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

ATM Transactions Per ATM (Robustness)

Table A1: Effect of Violence Shocks on Log ATM Transaction Intensity Per ATM

	Log ATM Transactions per ATM
Shock_t	-0.0145^{**}
	(0.00601)
Shock_{t-1}	-0.0115^{**}
	(0.00549)
Shock_{t-2}	-0.0104^{*}
	(0.00592)
Shock_{t-3}	-0.00789
	(0.00575)
Shock_{t-4}	-0.00414
	(0.00582)
Municipality×Calendar-Month FE	Yes
Year×Month FE	Yes
N	88,004

The dependent variable is the log of monthly ATM transactions per ATM. A municipality-month is classified as a shock if realized violence is at least twice the LSTM-predicted level. Shock_{t-k} denotes a violence shock occurring k months prior to period t, so that the coefficients capture the effect of a shock on the outcome up to k months after its occurrence. All five lags (from t to $t-4$) are included simultaneously in a distributed lag specification. Coefficients approximate percentage changes in ATM transaction intensity. Standard errors are clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OXXO Transactions Per Store (Robustness)

Table A2: Effect of Violence Shocks on Log OXXO Transaction Intensity Per Store

	Log OXXO Transactions per Store
Shock_t	0.00330
	(0.00613)
Shock_{t-1}	0.00361
	(0.00585)
Shock_{t-2}	0.00364
	(0.00547)
Shock_{t-3}	0.00373
	(0.00590)
Shock_{t-4}	0.00722
	(0.00607)
Municipality×Calendar-Month FE	Yes
Year×Month FE	Yes
N	49,717

The dependent variable is the log of monthly OXXO transactions per store. A municipality-month is classified as a shock if realized violence is at least twice the LSTM-predicted level. Shock_{t-k} denotes a violence shock occurring k months prior to period t, so that the coefficients capture the effect of a shock on the outcome up to k months after its occurrence. All five lags (from t to $t-4$) are included simultaneously in a distributed lag specification. Coefficients approximate percentage changes in OXXO transaction intensity. Standard errors are clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness to Violence Shock Threshold (ATM and OXXO)

Table A3: Robustness to Shock Threshold — ATM Transactions

	Panel A: Per Capita			*Panel B: Per ATM*		
	1.5×	1.75×	2×	1.5×	1.75×	2×
Shock_t	-0.0102^{**}	-0.0143^{***}	-0.0159^{***}	-0.00982^{**}	-0.0129^{**}	-0.0145^{**}
	(0.00470)	(0.00528)	(0.00580)	(0.00482)	(0.00533)	(0.00601)
Shock_{t-1}	-0.00743^{*}	-0.0103^{**}	-0.0129^{**}	-0.00810^{*}	-0.0104^{**}	-0.0115^{**}
	(0.00441)	(0.00486)	(0.00536)	(0.00442)	(0.00490)	(0.00549)
Shock_{t-2}	-0.00397	-0.00502	-0.00837	-0.00592	-0.00752	-0.0104^{*}
	(0.00443)	(0.00491)	(0.00546)	(0.00464)	(0.00528)	(0.00592)
Shock_{t-3}	-0.00216	-0.00255	-0.00566	-0.00392	-0.00473	-0.00789
	(0.00445)	(0.00509)	(0.00562)	(0.00450)	(0.00516)	(0.00575)
Shock_{t-4}	0.00136	0.00258	-0.00121	-0.0000738	-0.000421	-0.00414
	(0.00460)	(0.00495)	(0.00542)	(0.00472)	(0.00522)	(0.00582)
Mun × Cal-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × Month FE	Yes	Yes	Yes	Yes	Yes	Yes
N	87,962	87,962	87,962	88,004	88,004	88,004

All columns report the joint all-lags specification. Threshold refers to the minimum ratio of realized to LSTM-predicted violence required to classify a municipality-month as shocked. SE clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Robustness to Shock Threshold — OXXO Transactions

	Panel A: Per Capita			*Panel B: Per Store*		
	1.5×	1.75×	2×	1.5×	1.75×	2×
Shock_t	0.00368	−0.000532	−0.00153	0.00578	0.00113	0.00330
	(0.00428)	(0.00472)	(0.00534)	(0.00476)	(0.00542)	(0.00613)
Shock_{t-1}	0.00138	0.000269	−0.0000192	0.00318	0.00128	0.00361
	(0.00413)	(0.00429)	(0.00486)	(0.00466)	(0.00511)	(0.00585)
Shock_{t-2}	0.00185	0.00140	−0.00113	0.00390	0.00353	0.00364
	(0.00408)	(0.00432)	(0.00476)	(0.00460)	(0.00496)	(0.00547)
Shock_{t-3}	0.00198	−0.00112	−0.00141	0.00583	0.00253	0.00373
	(0.00391)	(0.00447)	(0.00472)	(0.00476)	(0.00543)	(0.00590)
Shock_{t-4}	0.000541	−0.000915	−0.000358	0.00671	0.00535	0.00722
	(0.00416)	(0.00463)	(0.00497)	(0.00489)	(0.00553)	(0.00607)
Mun × Cal-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × Month FE	Yes	Yes	Yes	Yes	Yes	Yes
N	55,378	55,378	55,378	49,717	49,717	49,717

All columns report the joint all-lags specification. Threshold refers to the minimum ratio of realized to LSTM-predicted violence required to classify a municipality-month as shocked. SE clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Low-Violence Placebo — All Four Outcomes

	Panel A: ATM Outcomes		*Panel B: OXXO Outcomes*	
	Per Capita	Per ATM	Per Capita	Per Store
$\text{Shock}_t^{\text{low}}$	0.00175	0.000521	−0.000820	0.000717
	(0.00527)	(0.00558)	(0.00793)	(0.00819)
$\text{Shock}_{t-1}^{\text{low}}$	0.00475	0.00214	0.00296	0.00444
	(0.00465)	(0.00507)	(0.00678)	(0.00683)
$\text{Shock}_{t-2}^{\text{low}}$	−0.0000291	−0.00122	0.00445	−0.000174
	(0.00515)	(0.00522)	(0.00678)	(0.00718)
$\text{Shock}_{t-3}^{\text{low}}$	0.00629	0.00397	0.000963	−0.00129
	(0.00481)	(0.00521)	(0.00596)	(0.00651)
$\text{Shock}_{t-4}^{\text{low}}$	0.00282	0.00129	−0.00324	−0.00452
	(0.00473)	(0.00501)	(0.00591)	(0.00723)
Mun × Cal-month FE	Yes	Yes	Yes	Yes
Year × Month FE	Yes	Yes	Yes	Yes
N	11,673	11,675	10,958	10,822

Placebo falsification using unexpectedly low-violence municipality-months, defined as realized violence at most half the LSTM-predicted level and strictly positive. The positivity restriction excludes months with mechanically zero violence that would otherwise satisfy the threshold by construction. SE clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 5: Average ATM Transactions per Capita by Municipality, 2014–2019

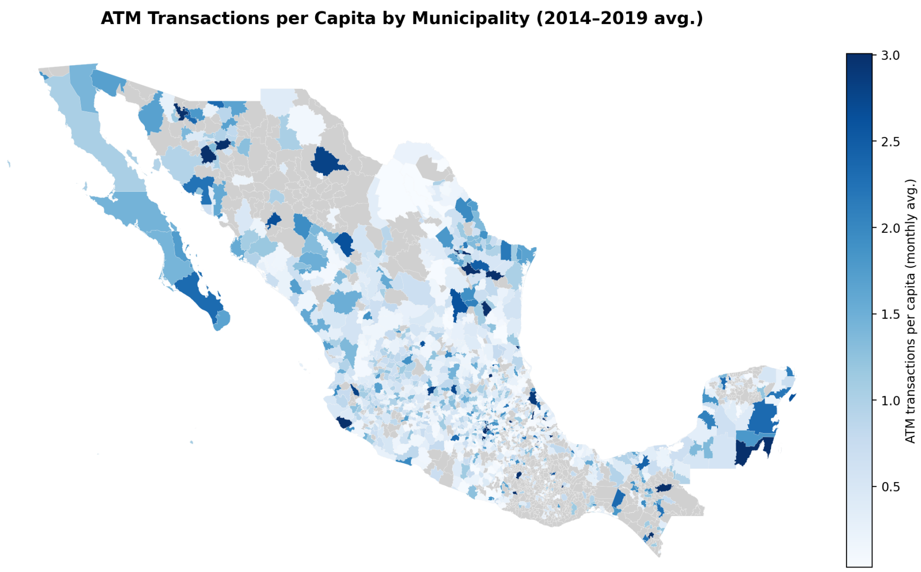


Figure 6: Average Number of OXXO Stores by Municipality, 2014–2019

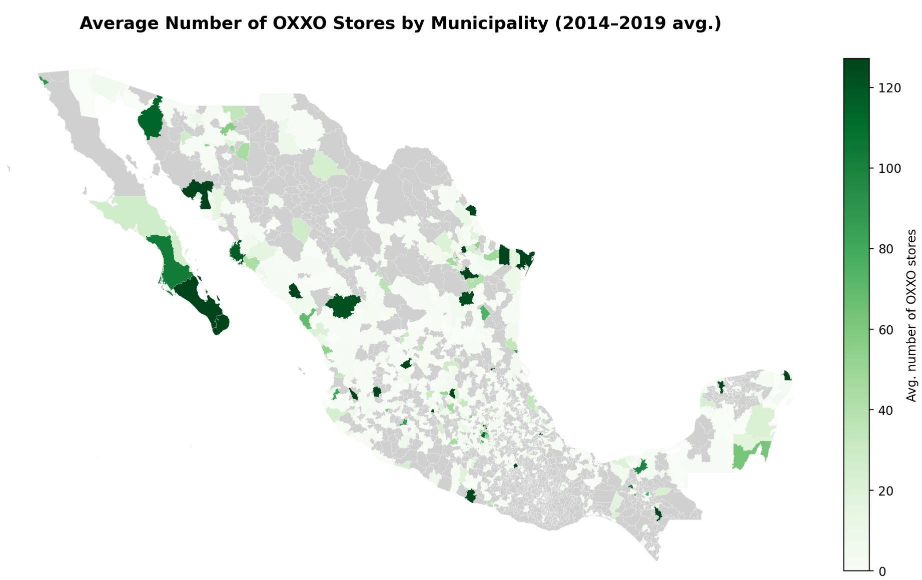


Figure 7: Municipality Coverage by Sample, ATM panel vs OXXO panel

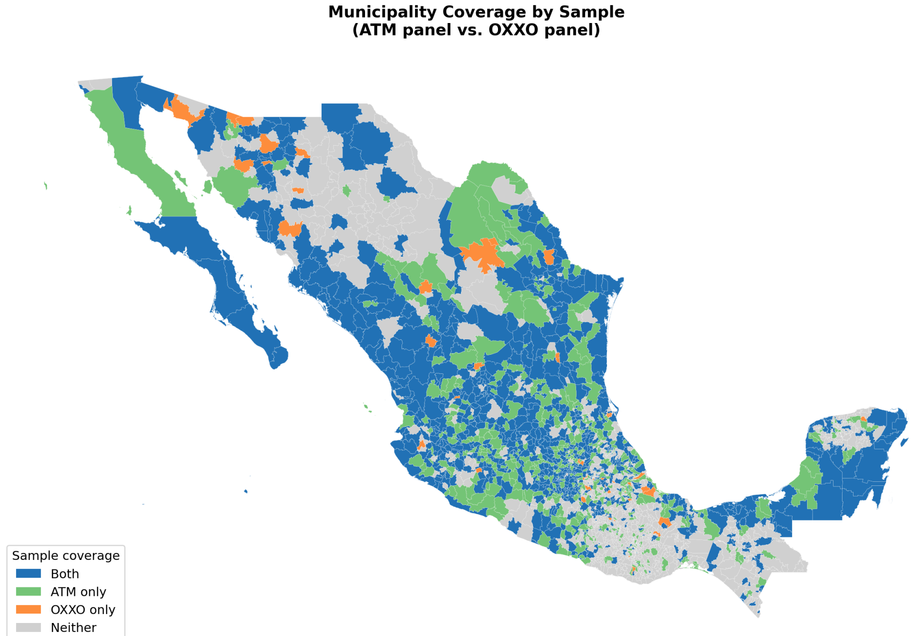


Figure 8: Event Study: ATM Transactions per Capita

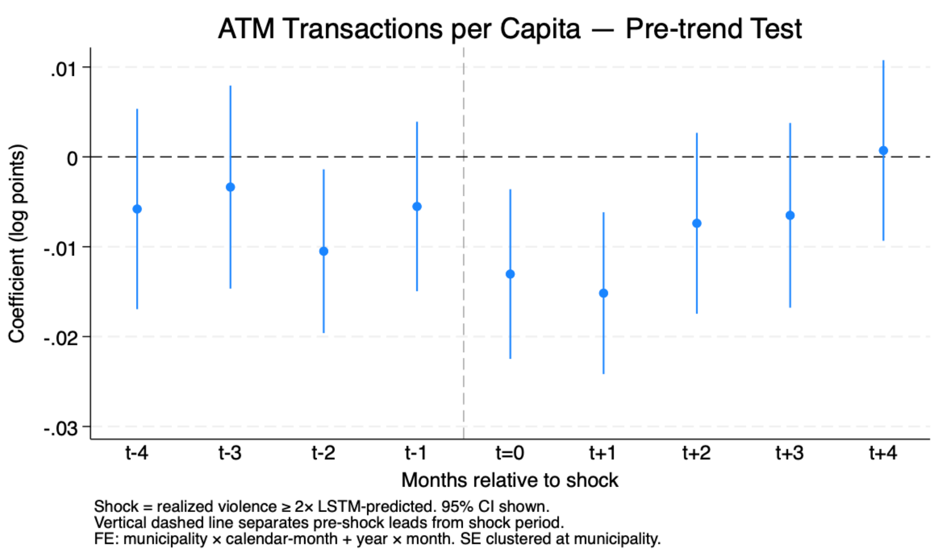


Figure 9: Event Study: ATM Transactions per ATM

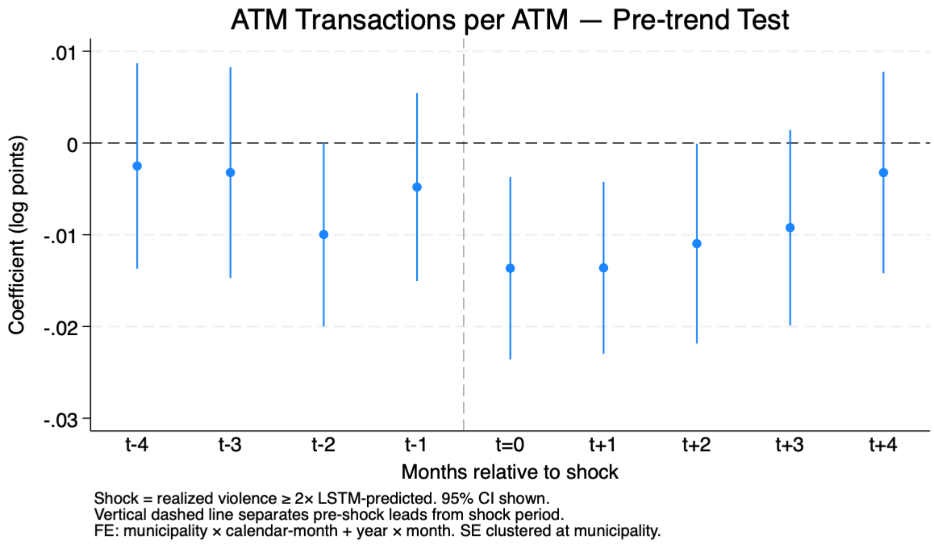


Figure 10: Event Study: OXXO Transactions per Capita

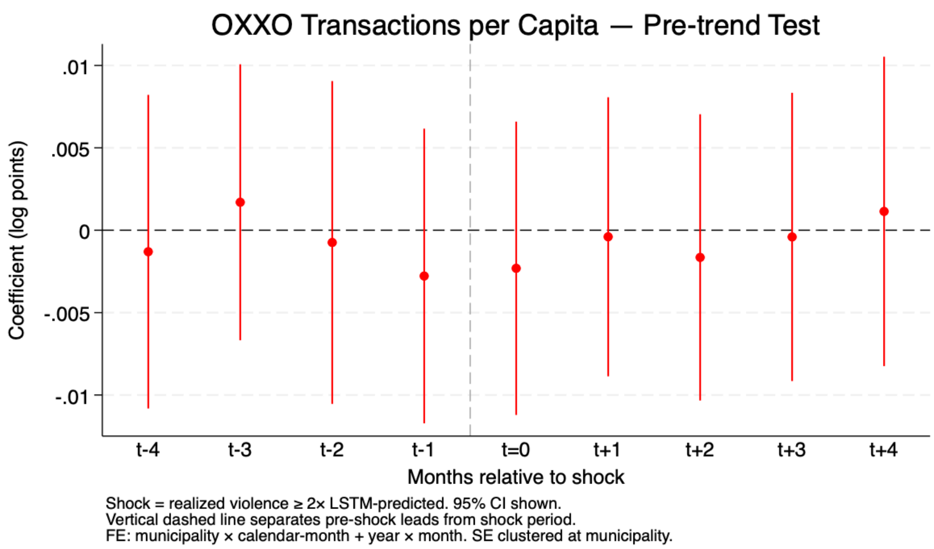
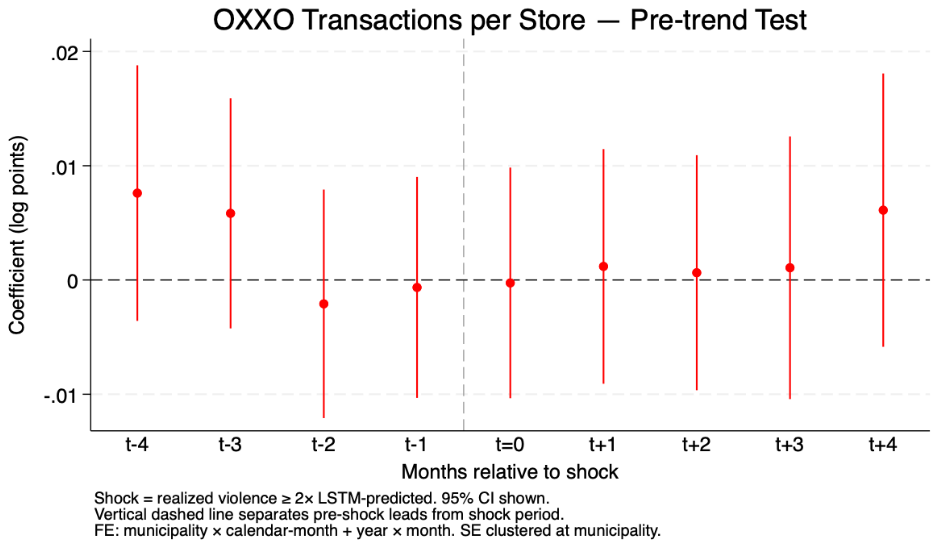


Figure 11: Event Study: OXXO Transactions per Store



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